Rule 606(a)(1) requires broker-dealers to produce quarterly public reports regarding their routing of non-directed orders[1090] in NMS stocks that are submitted on a held basis. Along with other information, these reports require the broker-dealer to report both the total dollar amount and per share average of net transaction fees paid and net transaction rebates received for different order types for each trading venue to which the broker-dealer reports routing orders.[1091] Additionally, Rule 606(b)(3) requires broker-dealers to produce reports pertaining to order handling upon the request of a customer that places, directly or indirectly, one or more orders in NMS stocks that are submitted on a not held basis, subject to a de minimis exception.[1092] For each venue to which the broker-dealer routed the customer's orders, these reports require the broker-dealer to disclose, among other things, the average net execution rebate or fee for shares of orders providing liquidity and the average net execution rebate or fee for shares of orders removing liquidity.[1093] However, these reports provide market participants with information only on historical average transaction fees and rebates and may not accurately reflect the current exchange fees and rebates a market participate will encounter at the time of its transaction.[1094]

1090 A "non-directed order" means any order from a customer other than a directed order. See 17 CFR 242.600(b)(56). A "directed order" means an order from a customer that the customer specifically instructed the broker or dealer to route to a particular venue for execution. See 17 CFR 242.600(b)(27).

1091 Rule 606(a)(1) requires broker-dealers to report separate information for market orders, marketable limit orders, non-marketable limit order, and other orders. See 17 CFR 242.606(a)(1) for the items that need to be disclosed in reports under rule 606(a)(1).

1092 See 17 CFR 242.606(b)(3). In addition, under rule 606(b)(5)'s customer-level de minimis exception, broker-dealers need not provide upon request execution quality reports for customers that traded on average each month for the prior six months less than $1,000,000 of notional value of not held orders in NMS stocks through the broker-dealer. See 17 CFR 242.606(b)(5).

1093 See 17 CFR 242.606(b)(3)(iii) and (iv).

1094 Reports under rule 606(a)(1) are produced by broker-dealers at the end of the quarter and disclose information on average fees and rebates for each month in that quarter. Reports issued by broker-dealers to their customers under rule 606(b)(3) disclose summarized information on the handling of the customer's orders for each calendar month over the prior six months. The broker-dealer must issue these reports to the customer within seven business days of receiving the customer's request.

c. Analysis of Current Access Fees and Rebates

The Commission analyzes, in Table 4, current fee and rebate schedules, based on Rule 19b-4 filings with the Commission, for each of the equity exchanges operating in the United States as of February 8, 2024,[1095] as well as the transaction prices that each exchange posts.[1096] What is apparent from this analysis is that the current structure of fees and rebates is complex and constantly changing.[1097] Each exchange, except LTSE which does not charge transaction fees, filed an average of 13.6 Rule 19b-4 equity market fee filings with the Commission in 2023. Market participants interacting with all exchanges had to adjust to 218 total fee filings in 2023. Each filing can contain changes for numerous fee and rebate categories.

The effect of the 30 mils fee cap as an anchor point is also apparent. For most exchanges the maximum fee assessed, presumably for non-protected quotes, is close to the 30 mils fee cap for protected quotes. The maximum rebate is generally in the vicinity of 30 mils, further suggesting the 30 mils access fee cap effectively limits what the exchanges offer as rebates. Some exchanges offer different access fees and rebate schedules for retail versus non-retail trades.[1098]

[1095] Table 4 is constructed using the same methodology as Table 5 of the Proposing Release, supra note 11, at 80311; the only difference is that Table 4 herein uses data as of Feb. 2024 and computes fee revisions during the 2023 calendar year, whereas Table 5 of the Proposing Release used data as of May 2022 and computed annual fee revisions from 2018 to June of 2022. Any differences between these two tables are due to changes in exchange fee schedules from May 2022 to Feb. 2024.

[1096] Panel A of Table 4 provides the category of exchange, maker-taker, inverted, or flat/free, the number of fee revisions since Jan. 2018 as indicated by the number of transaction fee specific rule 19b-4 filings that the exchange has filed with the Commission, the date that each exchange's website states that the fee schedule posted there is effective and the range of fees and rebates along with the number of categories of fees and rebates for transactions priced equal to, or greater than, $1.00 per share.

[1097] Some commenters have also stated that current transaction pricing practices introduce complexity into the market and reduce transparency. See Themis Letter at 7; BMO Letter at 3.

[1098] See, e.g., New York Stock Exchange (NYSE), Equity Fees and Charges, NYSE.COM, available at https://www.nyse.com/publicdocs/nyse/markets/nyse-arca/NYSE_Arca_Marketplace_Fees.pdf (accessed

Panel B of Table 4 provides information on the exchange's fee schedules for stocks priced lower than $1.00. For these transactions, the fee schedules tend to be simpler. Most exchanges do not offer a rebate for transactions lower than $1.00 even if the exchange offers rebates for other transactions – only three exchanges offer any sort of baseline rebate.[1099] Additionally, the exchanges tend to charge an access fee of 0.1% with some also charging the maximum access fee of 0.3% of the share price. Only one exchange charges a fee of 0.1% to both sides of a transaction.

Table 4: Summary of Transaction-Based Fee Schedules for U.S. National Equities Exchanges as of February 2024[a]

Panel A: Fees and Rebates for Transactions Greater than $1.00

Exchange	Fee Model	Number of Revisions 2023	Date of Fee Schedule	Fees (# of Categories)	Rebates (# of Categories)
Cboe BZX[b]	Maker-Taker	42	1/2/2024	$0.0030] (1)	[$0.0016-$0.0031 (7)
Cboe BYX[c]	Inverted	9	2/1/2024	$0.0012-$0.0020 (10)	$0.0015-$0.0020 (4)
Cboe EDGA[d]	Inverted	14	2/7/2024	$0.0000-$0.0030 (12)	$0.0016-$0.0024 (5)
Cboe EDGX[e]	Maker-Taker	44	2/1/2024	$0.00275-$0.0030 (3)	$0.0020-$0.0034 (8)
BX[f]	Inverted	6	2/1/2024	$0.0020-$-$0.0030 (2)	$0.0005-$0.0018 (7)
Phlx (PSX)[g]	Maker-Taker	4	2/1/2024	$0.0030 (1)	$0.0020-$0.0033 (5)
Nasdaq[h]	Maker-Taker	6	2/1/2024	$0.0030] (1)	$0.0013-$0.00305 (26)
	Maker-Taker	11	2/1/2024	$0.0000-$0.0030	$0.0000-$0.0032

June 18, 2024); see also Virtu Letter II at 23 suggesting that the Commission could consider an alternative which would allow exchanges to offer differing fee schedules to retail and non-retail orders.

1099 The three are Cboe EDGX, MEMX, and MIAX Pearl.

NYSE Arca[i]				(6)	(6)
NYSE American	Maker-Taker	7	1/3/2024	$0.0025-$0.0030 (3)	$0.0016-$0.0030 (3)
NYSE	Maker-Taker	16	1/12/2024	$0.0000-$0.00275 (5)	$0.0004-$0.0030 (11)
NYSE National	Inverted	5	1/3/2024	$0.0022-$0.0029 (4)	$0.0007-$0.0030 (5)
NYSE Chicago	Maker-Taker	3	1/8/2024	$0.0010-$0.0030 (1)	$0.0000 (0)
IEX[j]	Maker-Taker	2	1/24/2024	$0.0000-$0.0010 (2)	$0.0004 (1)
MEMX[k]	Maker-Taker	22	2/1/2024	$0.00295-$0.0030 (2)	$0.0015-$0.0033 (6)
MIAX Pearl[l]	Maker-Taker	10	1/17/2024	$0.0024 (1)	$0.00295 (1)
LTSE[m]	Free	NA	N/A	$0.0000 (1)	$0.0000 (1)

Panel B: Fees and Rebates for Transactions Under $1.00

Exchange	Fee Model	Rebate	Fee	Charged Both Sides
Cboe BZX	Maker-Taker	0	0.30%	
Cboe BYX	Inverted	0	0.10%	
Cboe EDGA	Inverted	0	0%	
Cboe EDGX	Maker-Taker	0.00009 (per share)	0.30%	
BX	Inverted	0	0.10%	
Phlx (PSX)	Maker-Taker	0	0.30%	
Nasdaq	Maker-Taker	0	0.30%	
NYSE Arca	Maker-Taker	0	0.10%	
NYSE American	Maker-Taker	0	0.10%	
NYSE	Maker-Taker	0	0.10%	
NYSE National	Inverted	0	0%	
NYSE Chicago	Maker-Taker	0	0.10%	Yes
IEX	Maker-Taker	0	0.09%	
MEMX	Maker-Taker	0.075% (of value)	0.10%	
MIAX Pearl	Maker-Taker	0.15%(of value)	0.250%	
LTSE	Free	0	0%	

[a] The number of fee revisions is obtained by counting each Rule 19b-4 filing for each exchange that is not clearly marked for a non-transaction fee related purpose such as connectivity fees, listing fees, options fees, etc. To determine the fee and rebate information, the staff searched each exchange's webpage for its current posted access fee and rebate schedule and collected information on access

fees and rebates pertaining to non-auction trading in stocks priced equal to, or greater than, $1.00 per share. Sources for Current Access Fee Data were effective on the dates shown in Panel A of Table 4, and were accessed during February 2024 at the websites shown beneath the table.

[b] https://www.cboe.com/us/equities/membership/fee_schedule/bzx/

[c] https://www.cboe.com/us/equities/membership/fee_schedule/byx/

[d] https://www.cboe.com/us/equities/membership/fee_schedule/edga/

[e] https://www.cboe.com/us/equities/membership/fee_schedule/edgx/

[f] https://www.nasdaqtrader.com/trader.aspx?id=bx_pricing

[g] https://www.nasdaqtrader.com/trader.aspx?id=psx_pricing

[h] https://www.nasdaqtrader.com/Trader.aspx?id=PriceListTrading2

[i] All NYSE Exchange Family fees: https://www.nyse.com/markets/fees

[j] https://exchange.iex.io/resources/trading/fee-schedule/ (Note: that the majority of IEX trading occurs via non-displayed orders. IEX only pays rebates on displayed orders.)

[k] https://info.memxtrading.com/equities-trading-resources/us-equities-fee-schedule/

[l] https://www.miaxglobal.com/sites/default/files/fee_schedule-files/MIAX_Pearl_Equities_Fee_Schedule_01172024.pdf

[m] https://ltse.com/trading/faqs

Complex fee schedules and volume-based tiers mean that it is difficult for the Commission to determine the net capture on a given exchange (the difference between average fees levied and rebates paid).[1100] Additionally, financial statements for exchange groups generally do not break down performance on a per-venue level and the financial statements generally combine auction access fees collected with regular trading access fees. Furthermore, some exchanges are privately held and thus do not release the same financial statements that public exchanges do. Using information from the financial statements of the three major exchange groups which collectively account for the overwhelming majority of trading volume on exchanges, the Commission estimates that the average total net capture is around 4 mils for all

1100 Volume-based tiers imply that the net capture varies by exchange member. To calculate an exchange's net capture would require knowing the number and types of orders that are executed by each member, mapping these orders to the exchange's fee schedule, calculating the net capture for each member, and then aggregating the net capture over all exchange members.

trading types.[1101] However, the Commission understands based on staff conversations with industry members that the net capture for non-auction trading in stocks that have a price equal to or greater than $1.00 is likely close to 2 mils on most exchanges,[1102] and in the analysis in later sections where the net capture needs to be assumed, we use 2 mils unless otherwise stated.[1103] A commenter agreed stating that the net fee is typically only 2 mils per share, stating, "it is worth pausing to reflect on just how competitive this net fee is—this is about as close to Bertrand price competition as one sees."[1104] Another commenter reported estimated trading related "all-in" costs to trade ranging 1.9 to 3.4 mils over the 2017 to 2021 period for the three largest exchange groups (Cboe, Nasdaq, and NYSE); [1105] these estimates are broadly in line with the estimated net-capture rates and consistent with the 2 mil net capture rate assumption used in the subsequent analysis.[1106] Given the low net capture rate of 2 mils on most exchanges, the primary reason that

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1101 Intercontinental Exchange, the parent firm of NYSE, reports on page 53 of its 2021 Form 10-K filing that their net capture for U.S. equity transactions was approximately 4.2 mils in 2021. Nasdaq did not report its net capture in its Form 10-K filing, however Nasdaq provides information on its investor relations webpage which, when we average the relevant 2021 volumes, indicates that the average net capture across all Nasdaq platforms for U.S. equity transactions was 5.9 mils. See Nasdaq 2022/2021 Monthly Volumes, Nasdaq, available at https://ir.nasdaq.com/static-files/465d2157-c476-4546-a9f7-8d7ad0c9be77). Cboe reports in its Form 10-K filing that its net capture for U.S. equity transactions was approximately 2 mils.

1102 Non-auction orders exclude opening, closing, and reopening auctions. See Table 7, note a for additional details regarding which orders are considered for estimation.

1103 One commenter stated that exchanges subsidize rebates with other sources of revenue as manifest by the fact that some market participants could receive rebates in excess of the 30 mil fee cap. See Healthy Markets Letter I at 23. Table 4 presents evidence consistent with the notion that in some cases rebates received may be in excess of 30 mils. However, the commenter did not provide any analysis to suggest that the net capture of the exchanges was, on average, negative. As discussed, the Commission believes that most exchanges, on average, earn approximately 2 mils per transaction priced greater than $1.00. One exception is IEX which earns an estimated 6 mils based on the information in Table 4.

1104 See Budish Letter at 3. Bertrand competition is an economics model of competition on the basis of prices whereby firms set their price—net price in this instance—at marginal cost.

1105 The commenter also included similar estimates for IEX which ranged from 6.8-8.9 mils, see Nasdaq Letter II at 3.

1106 Subsequent analysis in section VII.D.2 assumes that the assumed 2 mil net capture will continue to be valid following the implementation of the amendments.

access fees remain near 30 mils on most exchanges is likely to fund rebates.[1107] For stocks trading below $1.00 the Commission estimates an average net capture of around 0.24% of the transaction volume.[1108] This amount is close to the 0.30% access fee cap and arises because, as seen in Panel B of Table 4, much of the trading for sub $1.00 priced volume takes place on exchanges which set their baseline fee at or near 0.30% but do not offer baseline rebates for transactions under $1.00. On a per-share basis, this net capture of 0.24% of transaction dollar volume corresponds to a net capture of 7.3 mils.[1109]

Table 5 presents tabulations of the total share (Panel A) and dollar (Panel B) trading volume executed on the 16 exchanges in 2023.[1110] This table provides estimates for the total volume that executed below $1.00, and that which executed above $1.00. These numbers represent an estimate of the total number of shares that will have been subject to the access fees and rebates discussed in this release.

Table 5: Trading volume by Exchange, Exchange Type 2023[a]

Panel A: Share Volume

Exchange Name	Exchange Type	<$1 Volume (Billions)	>=$1 Volume QS<= $0.015 (Billions)	>=$1 Volume QS > $0.015 (Billions)	% of Exchange Volume

[1107] See Retirement Coalition Letter at 1 ("in practice, this 'cap' has come to be used as the standard rate charged to access quotes at most exchanges, and almost all of those fees are then 'rebated' to liquidity providers").

[1108] The estimate for the 0.24% net capture is obtained by taking the total estimated net transaction fee across all exchanges for trading in shares priced below $1.00 ($83.2 million) and dividing this number by the total sub $1.00 dollar volume from Panel B of Table 5 below ($34.2 billion). 100*83.2million/34.2 billon ≈%0.24.

[1109] To estimate the net capture in terms of mils, the Commission divides the dollar revenue from fees by the number of shares traded. The dollar net capture is 0.24%, see id.,) and the dollar volume is $34.2 billion, see Table 5, Panel B, resulting in a dollar revenue of $82 million (0.0024*$34.2 billion). With share volume of 112.6 billion, see Table 5, Panel A, the net capture is 7.3 mils ($82 million / 112.6 billion).

[1110] Table 5 is constructed using the same methodology as Table 6 of the Proposing Release, supra note 11, at 80313; Table 5 herein uses data for 2023, whereas Table 6 of the Proposing Release used data for the first half of 2022.

Off-Exchange		192.9	620.3	241.3	
Nasdaq	Maker-Taker	26.8	226.2	88.9	26.9%
NYSE Arca	Maker-Taker	27.5	139.3	30.4	15.5%
NYSE	Maker-Taker	3.4	127.4	37.4	13.2%
Cboe BZX	Maker-Taker	14.0	86.3	20.9	9.5%
Cboe EDGX	Maker-Taker	21.1	98.2	23.3	11.2%
MEMX	Maker-Taker	8.1	61.9	12.0	6.5%
IEX	Maker-Taker	1.7	38.9	19.5	4.7%
Cboe EDGA	Inverted	2.7	33.8	5.6	3.3%
Cboe BYX	Inverted	2.6	20.4	2.8	2.0%
MIAX Pearl	Maker-Taker	2.4	41.9	2.7	3.7%
NYSE National	Inverted	0.5	11.0	1.4	1.0%
Nasdaq OMX PSX	Maker-Taker	0.3	7.5	1.9	0.8%
Nasdaq OMX BX	Inverted	0.5	6.7	2.6	0.8%
NYSE American	Maker-Taker	1.0	4.8	1.0	0.5%
NYSE Chicago	Flat	0.2	0.8	1.8	0.2%
LTSE	Flat	0.0	0.0	0.0	0.0%
	Total	305.5	1,525.4	493.5	
	Exchange Total	112.6	905.1	252.2	

Panel B: Dollar Volume

Exchange Name	Exchange Type	<$1 Volume (Billions)	>=$1 Volume QS<= $0.015 (Billions)	>=$1 Volume QS > $0.015 (Billions)	% of Exchange Volume
Off-Exchange	-	65.3	19,744.9	24,508.4	
Nasdaq(TapeC)	Maker-Taker	9.1	9,008.3	9,402.3	30.4%
NYSE Arca	Maker-Taker	7.6	6,433.3	3,107.1	15.8%
NYSE	Maker-Taker	1.5	3,985.9	3,658.6	12.6%
Cboe BZX	Maker-Taker	3.1	3,711.6	2,479.9	10.2%
Cboe EDGX	Maker-Taker	6.2	3,450.0	2,371.1	9.6%
MEMX	Maker-Taker	2.5	2,148.8	1,130.4	5.4%
IEX	Maker-Taker	0.8	1,383.8	2,119.0	5.8%
Cboe EDGA	Inverted	1.0	1,038.0	494.8	2.5%
Cboe BYX	Inverted	0.9	657.2	275.3	1.5%
MIAX	Maker-Taker	0.7	1,297.3	268.4	2.6%
NYSE National	Inverted	0.2	287.3	131.0	0.7%
Nasdaq OMX PSX	Maker-Taker	0.1	368.9	211.9	1.0%
Nasdaq OMX BX	Inverted	0.2	274.9	258.7	0.9%
NYSE American	Maker-Taker	0.4	165.5	97.7	0.4%
NYSE Chicago	Flat	0.1	39.6	236.0	0.5%
LTSE	Flat	0.0	1.5	1.6	0.0%

Total	99.4	53,996.9	50,752.2
Exchange Total	34.2	34,252.0	26,243.9

[a] This table aggregates all trade information from the TAQ database for every trading day in 2023. Only trading volume reflecting normal trades during regular trading is included. Normal trades are identified in TAQ data by sale conditions "blank, @, E, F, I, S, Y" which correspond to regular trades, intermarket sweep orders, odd-lot trades, split trades, and yellow flag regular trades. The remaining share volume was aggregated by exchange, and the table denotes exchange type (maker-taker, inverted, flat, free). Share and dollar volume from exchange codes T and Q were combined into 'Nasdaq.' Panel A presents share volume totals and Panel B presents dollar volume totals. Certain items in Table 5 may also be affected by the MDI Rules once they are fully implemented. See infra section VII.C.3.

Transaction fees for trades in stocks priced equal to or greater than $1.00 are generally levied per share transacted. From Table 5 we see that in 2023, there were approximately 2 trillion shares transacted at prices equal to or greater than $1.00 per share across all venues, 57% of which (1.16 trillion shares) were executed on a registered exchange.[1111] Of these on-exchange transactions priced equal to or greater than $1.00 per share, approximately 78% were in stocks with quoted spreads of $0.015 or less.[1112] These numbers provide the basis for estimating the total amount of access fees and rebates collected and distributed in transactions priced equal to, or greater than, $1.00 per share. For transactions less than $1.00 per share the access fee is generally levied as a percent of the transaction share price. In Panel B we see that in 2023 there was approximately $34 billion transacted on exchanges in shares priced less than $1.00 per share.

1111 2T shares ≈ 1.5T narrow spread shares + 493 billion wider spread shares. Also, off-exchange trading volume has increased in recent years. See, e.g., Jonathan Brogaard & Jing Pan, Dark Pool Trading and Information Acquisition, 35 REV. FIN. STUD. 2625 (2022).

1112 The fourth column of Panel A shows 905.1 billion shares traded on exchanges with a price greater than or equal to $1.00 and a quoted spread of $0.015 or less; the fifth column shows 252.2 billion shares traded on exchanges with a price greater than or equal to $1.00 and a quoted spread over $0.015. The total number of shares traded on exchanges with a price greater than or equal to $1.00 is therefore 1157.3 billion (905.1+252.2), and the fraction of these that a spread of $0.015 or less is 78% (905.1/1157.3).

Panels A and B of Table 6 break down the share and dollar volume statistics presented in Table 5 by venue type: maker-taker, inverted, and flat/free.[1113] The overwhelming majority (over 90%) of both dollar and share exchange trading volume occurs on maker-taker venues. Inverted exchanges capture about 5-7% of dollar and share volume, and the remaining share volume transact on flat/free exchanges.

Table 6: Volume by Exchange Type and Estimated Access Fee/Rebate Estimates 2023[a]

Panel A: Exchange Share Volume By Venue Type

	Price<\$1 (Billions)	Price>\$1; TWAQS ≤ \$0.015 (Billions)	Price>\$1; TWAQS > \$0.015 (Billions)	% Total
Maker-Taker	106.2	832.4	237.9	92.7%
Inverted	6.2	71.9	12.4	7.1%
Flat/Free	0.2	0.8	1.8	0.2%

Panel B: Exchange Dollar Volume by Venue Type

	Price<\$1 (Billions)	Price>\$1; TWAQS ≤ \$0.015 (Billions)	Price>\$1; TWAQS > \$0.015 (Billions)	% Total
Maker-Taker	31.9	31,953.4	24,846.4	93.9%
Inverted	2.2	2,257.4	1,159.8	5.6%
Flat/Free	0.1	41.1	237.6	0.5%

Panel C: Estimated Fees Collected and Rebates Distributed (Billions)

Fees Collected	\$3.41
Rebates Distributed	\$3.08
Exchange Capture	\$0.34

Panel D: Total Estimated Net Fees by Liquidity Type (Billions)

Demander	\$2.97
Provider	(\$2.63)
Exchange Capture	\$0.34

[a] Certain items in this Table 6 may also be affected by the amendments in the MDI Rules once they are fully implemented. See infra section VII.C.3.

[1113] Table 6 is constructed using the same methodology as Table 7 of the Proposing Release, supra note 11, at 80314. The only difference is that Table 6 herein uses data for 2023, whereas Table 7 of the Proposing Release used data for the first six months of 2022.